

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Computr Engg/ CSE Semester :IV Subject Code & Name: BTES405 Digital Logic Design & Microprocessor Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Differentiate between analog vs digital signal.	Analyzing	6
B)	Which gates are known as universal gates? Justify using examples.	Understanding	6
C)	State and prove any two theorems of Boolean algebra.	Applying	6
Q.2	Solve Any Two of the following.		12
A)	How will you implement Full adder circuit? Draw the circuit diagram and derive equation for sum and carry.	Understanding	6
B)	Using K map, simplify Boolean equation for the following logic equation expressed by min terms? $Y=F(A,B,C,D)=\sum m(7,9, 10, 11, 12, 13, 14, 15)$	Applying	6
C)	Differentiate between combinational and sequential logic circuit.	Analyzing	6
Q. 3	Solve Any Two of the following.		12
A)	Differentiate between synchronous and asynchronous counter.	Analyzing	6
B)	Explain SR Flip flop in detail.	Understanding	6
C)	Draw and explain serial in serial out shift register in detail.	Understanding	6
Q.4	Solve Any Two of the following.		12
A)	Differentiate in between 8085 & 8086 microprocessors.	Analyzing	6
B)	Draw & explain architecture of DMA controller.	Understanding	6
C)	Draw & explain 8086 block diagram.	Understanding	6
Q. 5	Solve Any Two of the following.		12
A)	Classify different instruction set of 8086.	Analyzing	6

B)	Explain different addressing modes of 8086.	Understanding	6
C)	Explain assembly language programming tools.	Understanding	6
	*** End ***		